$$\mathcal{C}(\tau) = \sum_{ij} c_i c_j \left\langle \hat{q}_i(\tau)\hat{q}_j \right\rangle_{\hat{\rho}_\mathcal{E}} = \sum_i c_i^2 \left\langle \hat{q}_i(\tau)\hat{q}_i \right\rangle_{\hat{\rho}_\mathcal{E}} . \tag{5.6}$$